

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1

PAPER 1 November 2021

2 hours 30 minutes

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2021 4004/1.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. Work out $2/5 + 2/3$. Give a mixed number if possible. **[3]**

2. Evaluate $9 + 4 \times 2 - 3$. **[2]**

3. Overnight in Masvingo the temperature was 0°C . By 14:00 it was 20°C . Find the rise. **[2]**

4. A map of Masvingo uses scale 2 : 500000. A road is 10 cm on the map. Find the real length in km. (1 : 100 000 means 1 cm $\leftrightarrow$ 1 km if 2:5 is first simplified — instead: scale 2 cm to 5 km.) **[3]**

5. In a class of St Mary's, $n(A) = 11$ (play soccer), $n(A \cap B) = 5$ (play both). How many play soccer only? **[2]**

6. Find 25% of 80. **[3]**

7. Expand and simplify $6(x + 3) - 5x$. **[3]**

8. Write 80000 in standard form. **[3]**

- 9.** Simplify $(3^4)^4$. Leave as a power of 3. **[3]**
- 10.** Find the HCF and LCM of 12 and 28. **[4]**
- 11.** Write the equation of the line with gradient 3 passing through (0, 5). **[3]**
- 12.** The n th term is $6 + 4(n - 1)$. Find the 8th term. **[3]**
- 13.** A regular polygon has 5 sides. Find each interior angle. **[4]**
- 14.** Factorise $x^2 + 7x + 6$. **[4]**
- 15.** Make h the subject of $A = 2\pi r(r + h)$. **[4]**
- 16.** In triangle ABC, $\hat{A} = 39^\circ$ and $\hat{B} = 57^\circ$. Find $\hat{C}$. **[3]**
- 17.** Find the midpoint of (3, 7) and (3, 3). **[3]**
- 18.** A triangular plot has base 6 m and perpendicular height 5 m. Find its area. **[3]**
- 19.** Factorise $x^2 - 4$. **[4]**
- 20.** $u = (1 ; 1)$, $v = (5 ; 5)$. Find $2u + v$. **[4]**

- 21.** A bag contains 5 red, 3 blue and 1 green beads. $P(\text{not green})$? **[3]**
- 22.** A ladder of length 5 m leans against a wall. The foot is 3 m from the wall. How far up the wall does it reach? **[4]**
- 23.** Find the mean of 7, 6, 9, 10, 7. **[3]**
- 24.** Area of a circle radius 21 cm. Take $\pi = 22/7$. **[4]**
- 25.** A cylindrical drum has radius 7 cm and height 6 cm. Find its volume. $\pi = 22/7$. **[4]**
- 26.** A shirt in Masvingo is marked \$200. In a sale it is 25% off. Sale price? **[4]**
- 27.** Solve $2(x + 4) = 18$. **[4]**
- 28.** Solve $2x + 1 < 5$. **[3]**
- 29.** In right-angled triangle PQR, $P = 30^\circ$ and hypotenuse $PR = 16$ cm. Find PQ, opposite 30° . **[4]**
- 30.** Solve
$$\begin{cases} x + y = 5 \\ x - y = 3 \end{cases}$$
 [4]

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).